

## On a problem of Erdős related to common factor differences

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Let  $n$  be a positive integer. The factor-difference set  $D(n)$  of  $n$  is the set of absolute values  $d$  of the differences between the factors of any factorization of  $n$  as a product of two integers. Erdős and Rosenfeld [The factor-difference set of integers, *Acta Arith.* **79**(4) (1997) 353–359] ask whether for every positive integer  $n$  there exist integers  $N_1 < \dots < N_n$  such that  $|\bigcap_{i=1}^n D(N_i)| \geq n$ , and prove this is true when  $n = 2$ . Urroz [A note on a conjecture of Erdős and Rosenfeld, *J. Number Theory* **78**(1) (1999) 140–143] shows the result true for  $n = 3$ . The ideas of this paper can be extended, and here, we show the result true for  $n = 4$  by proving there are infinitely many sets of four integers with four common factor differences.

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### 1. Introduction

Erdős observed that the maximum number of odd integral distances determined by  $3n$  points in the plane is  $3n^2$ , and asked whether the bound could be obtained (the answer is yes, by a result of Piepmeyer [3]). In investigating the problem, Erdős and Rosenfeld [2] supposed  $2n$  points  $K_i(\pm(2k_i + 1)/2, 0)$  on the  $x$ -axis, for integers  $k_i$ , and hoped to find  $n$  points  $P_j(0, p_j)$  on the  $y$ -axis,  $p_j \in \mathbb{Z}$ , such that all distances  $d_{i,j}$  between  $K_i$  and  $P_j$  were odd integers. This requires

$$4p_j^2 = (2d_{i,j} - (2k_i + 1))(2d_{i,j} + (2k_i + 1)),$$

and each  $4p_j^2$ ,  $j = 1, \dots, n$ , contains  $4k_i + 2$ ,  $i = 1, \dots, n$ , as difference between factors. If we define the factor-difference set  $D(n)$  by

$$D(n) = \{d : d = |a - b|, ab = n\},$$

then the above construction leads to asking whether for every positive integer  $n$  there exist integers  $N_1 < \dots < N_n$  such that  $|\bigcap_{i=1}^n D(N_i)| \geq n$ . They prove this result holds when  $n = 2$ . (In fact, they show that for all positive integers  $k$ ,

there exist integers  $N_i$ ,  $i = 1, \dots, k$  with  $|\bigcap_{i=1}^k D(N_i)| \geq 2$ .) Urroz [4] (see also Bremner [1], the review of this paper in Mathematical Reviews by the current author) shows a similar result true for  $n = 3$ . The ideas of this paper can be extended, and here, we show the result true for  $n = 4$  by proving there are infinitely many sets of four integers with four common factor differences.

For the case  $n = 4$ , suppose that  $N_i$ ,  $i = 1, \dots, 4$ , are four integers possessing four common factor-differences  $2x_0$ ,  $2y_0$ ,  $2z_0$ ,  $2t_0$ . Then we obtain a system of 12 quadratic equations:

$$(N_i =) x_i^2 - x_0^2 = y_i^2 - y_0^2 = z_i^2 - z_0^2 = t_i^2 - t_0^2, \quad i = 1, \dots, 4. \quad (1.1)$$

A solution of this system is said to be non-degenerate if the  $N_i$  are distinct and non-zero. For example, such a solution is given by the four integers

$$(N_1, N_2, N_3, N_4) = (26128575, 291722431, 561117375, 713526975), \quad (1.2)$$

with common factor differences

$$(x_0, y_0, z_0, t_0) = (63, 8055, 16695, 37695). \quad (1.3)$$

Here, the  $x_i, y_i, z_i, t_i$  for  $i = 1, \dots, 4$ , can be computed from (1.1), and equal

$$\begin{aligned} (x_1, y_1, z_1, t_1) &= (5112, 9540, 17460, 38040), \\ (x_2, y_2, z_2, t_2) &= (17080, 18884, 23884, 41384), \\ (x_3, y_3, z_3, t_3) &= (23688, 25020, 28980, 44520), \\ (x_4, y_4, z_4, t_4) &= (26712, 27900, 31500, 46200). \end{aligned}$$

(Note: We shall always order numerical solutions so that  $0 < x_0 < y_0 < z_0 < t_0$ .) The solution (1.2) trivially gives rise to an infinity of solutions simply by multiplying by  $d^2$  for any integer  $d$ ; but our interest is in constructing infinitely many solutions with distinct ratios  $N_1 : N_2 : N_3 : N_4$ . Over many weeks we computed 211 distinct solutions in integers to the system (1.1), the first few of which, ordered by the sum  $N_1 + N_2 + N_3 + N_4$ , are listed in Appendix A. The above solution at (1.2), (1.3), is the one found with minimal  $x_0$ . It seems highly likely that there should exist a parametric solution to the system (1.1), but despite extensive efforts we have (as yet) been unable to find such. However, it is shown in this paper that there are infinitely many distinct integer solutions to (1.1), with consequence that there are infinitely many quadruples of integers, each sharing four common factor-differences.

## 2. The Construction

The system (1.1) is equivalent to

$$x_0^2 - y_0^2 = x_i^2 - y_i^2, \quad x_0^2 - z_0^2 = x_i^2 - z_i^2, \quad x_0^2 - t_0^2 = x_i^2 - t_i^2, \quad i = 1, \dots, 4. \quad (2.1)$$

First, consider the equations

$$x_0^2 - y_0^2 = x_1^2 - y_1^2 = x_2^2 - y_2^2; \quad x_0^2 - z_0^2 = x_1^2 - z_1^2 = x_2^2 - z_2^2.$$

We may set

$$\begin{aligned} x_0 + y_0 &= b, & x_1 + y_1 &= c, & x_2 + y_2 &= d, \\ x_0 - y_0 &= mcd, & x_1 - y_1 &= mbd, & x_2 - y_2 &= mbc, \\ x_0 + z_0 &= pq, & x_1 + z_1 &= pr, & x_2 + z_2 &= ps, \\ x_0 - z_0 &= lrs, & x_1 - z_1 &= lqs, & x_2 - z_2 &= lqr; \end{aligned}$$

so that

$$\begin{aligned} 2(x_0, x_1, x_2) &= (b + mcd, c + mbd, d + mbc) \\ &= (pq + lrs, pr + lqs, ps + lqr), \\ 2(y_0, y_1, y_2) &= (b - mcd, c - mbd, d - mbc), \\ 2(z_0, z_1, z_2) &= (pq - lrs, pr - lqs, ps - lqr). \end{aligned}$$

The compatibility conditions

$$b + mcd = pq + lrs, \quad c + mbd = pr + lqs, \quad d + mbc = ps + lqr$$

may be solved for  $l, m, p$  to give

$$\begin{aligned} (l, m, p) &= \frac{1}{\Delta} (b(c^2 - d^2)q + c(d^2 - b^2)r + d(b^2 - c^2)s, bq(s^2 - r^2) + cr(q^2 - s^2) \\ &\quad + ds(r^2 - q^2), -(dqr(b^2 - c^2) + brs(c^2 - d^2) + csq(d^2 - b^2))), \end{aligned}$$

where

$$\Delta = bc(q^2 - r^2)s + cd(r^2 - s^2)q + db(s^2 - q^2)r,$$

from which we get the following formulae for  $x_0, y_0, z_0; x_1, y_1, z_1; x_2, y_2, z_2$ , in terms of the parameters  $b, c, d, q, r, s$ .

$$\begin{aligned} 2\Delta x_0 &= -(d^2 - b^2)c(q^2 - r^2)s + (b^2 - c^2)d(s^2 - q^2)r, \\ 2\Delta y_0 &= (b^2 + d^2)c(q^2 - r^2)s + 2bcd(r^2 - s^2)q + (b^2 + c^2)d(s^2 - q^2)r, \\ 2\Delta z_0 &= -(d^2 - b^2)c(q^2 + r^2)s - (b^2 - c^2)d(s^2 + q^2)r - 2(c^2 - d^2)bqrs; \\ 2\Delta x_1 &= (c^2 - d^2)b(q^2 - r^2)s - (b^2 - c^2)d(r^2 - s^2)q, \\ 2\Delta y_1 &= (c^2 + d^2)b(q^2 - r^2)s + (b^2 + c^2)d(r^2 - s^2)q + 2bcd r(s^2 - q^2), \\ 2\Delta z_1 &= -(c^2 - d^2)b(q^2 + r^2)s - (b^2 - c^2)d(r^2 + s^2)q - 2cqr s(d^2 - b^2); \\ 2\Delta x_2 &= (d^2 - b^2)c(r^2 - s^2)q - (c^2 - d^2)b(s^2 - q^2)r, \\ 2\Delta y_2 &= 2bcd(q^2 - r^2)s + (b^2 + d^2)c(r^2 - s^2)q + (c^2 + d^2)b(s^2 - q^2)r, \\ 2\Delta z_2 &= -(c^2 - d^2)b(s^2 + q^2)r - (d^2 - b^2)c(r^2 + s^2)q - 2(b^2 - c^2)dqrs. \end{aligned} \tag{2.2}$$

Set  $(A, B) = (x_0^2 - z_0^2, x_0^2 - y_0^2)$ , and consider the elliptic curve

$$E_1 : V^2 = U(U - A)(U - B).$$

This curve contains the points  $P_0(x_0^2, x_0y_0z_0)$ ,  $P_1(x_1^2, x_1y_1z_1)$ , and  $P_2(x_2^2, x_2y_2z_2)$ . Moreover, since  $x_i^2$ ,  $x_i^2 - A$ ,  $x_i^2 - B$ ,  $i = 0, 1, 2$ , are all squares, it follows that the points  $P_i$  all lie in  $2E_1(\mathbb{Q})$ . We let  $x_3^2$  be the  $U$ -coordinate of any point in  $2E_1(\mathbb{Q})$  other than  $P_0, P_1, P_2$ , and then necessarily  $x_3^2 - A$  and  $x_3^2 - B$  will be squares, and  $y_3, z_3$  are determined as rationals. Similarly, for  $x_4^2$ . Take, for example,

$$x_3^2 = U(P_1 + P_2) = \left( \frac{x_1y_2z_2 - x_2y_1z_1}{x_1^2 - x_2^2} \right)^2,$$

and

$$x_4^2 = U(P_0 + P_1) = \left( \frac{x_0y_1z_1 - x_1y_0z_0}{x_0^2 - x_1^2} \right)^2.$$

Then

$$(x_3, y_3, z_3) = \left( \frac{x_1y_2z_2 - x_2y_1z_1}{x_1^2 - x_2^2}, \frac{y_1x_2z_2 - y_2x_1z_1}{y_1^2 - y_2^2}, \frac{z_1y_2x_2 - z_2y_1x_1}{z_1^2 - z_2^2} \right);$$

$$(x_4, y_4, z_4) = \left( \frac{x_0y_1z_1 - x_1y_0z_0}{x_0^2 - x_1^2}, \frac{y_0x_1z_1 - y_1x_0z_0}{y_0^2 - y_1^2}, \frac{z_0y_1x_1 - z_1y_0x_0}{z_0^2 - z_1^2} \right).$$

We now have  $(N_1, N_2, N_3, N_4) = (x_1^2 - x_0^2, x_2^2 - x_0^2, x_3^2 - x_0^2, x_4^2 - x_0^2)$  given by rational functions of  $b, c, d, q, r, s$ . It remains to find  $t_0, \dots, t_4$  with

$$t_0^2 + N_1 = t_1^2, \quad t_0^2 + N_2 = t_2^2, \quad t_0^2 + N_3 = t_3^2, \quad t_0^2 + N_4 = t_4^2. \quad (2.3)$$

The elliptic curve

$$E_2 : Y^2 = X(X + N_1)(X + N_2)$$

contains the points  $Q_x(x_0^2, x_0x_1x_2)$ ,  $Q_y(y_0^2, y_0y_1y_2)$ , and  $Q_z(z_0^2, z_0z_1z_2)$ , which all lie in  $2E_2(\mathbb{Q})$ . We can take  $t_0^2$  to be the  $X$ -coordinate of any point  $Q_t$  in  $2E_2(\mathbb{Q})$  other than  $Q_x, Q_y, Q_z$ , and then

$$t_0^2 + N_1 = t_1^2, \quad t_0^2 + N_2 = t_2^2.$$

We choose

$$t_0^2 = X(Q_x - Q_z) = \left( \frac{x_1x_2z_0 + x_0z_1z_2}{x_0^2 - z_0^2} \right)^2.$$

Then

$$(t_0, t_1, t_2) = \left( \frac{x_1x_2z_0 + x_0z_1z_2}{x_0^2 - z_0^2}, \frac{x_2x_0z_1 + x_1z_2z_0}{x_1^2 - z_1^2}, \frac{x_0x_1z_2 + x_2z_0z_1}{x_2^2 - z_2^2} \right). \quad (2.4)$$

Of the conditions at (2.3), it remains to satisfy

$$t_0^2 + N_3 = t_3^2, \quad t_0^2 + N_4 = t_4^2.$$

It seems that an appropriate strategy now would be to consider the curve

$$Y^2 = X(X + N_3)(X + N_4),$$

and argue as in the preceding to obtain  $t'_0$  with  $t'^2_0 + N_3 = \square$  and  $t'^2_0 + N_4 = \square$ . Then we obtain a single condition for a solution to our problem by equating  $t^2_0 = t'^2_0$ . But this approach seems to lead to intractable polynomials; and thus we resort to a different curve. Consider

$$E_3 : Y^2 = X(X - N_3)(X - (N_3 - N_2)) = X(X - (x^2_3 - x^2_0))(X - (x^2_3 - x^2_2)),$$

with points

$$R_x(x^2_3, x_0x_2x_3), \quad R_y(y^2_3, y_0y_2y_3), \quad R_z(z^2_3, z_0z_2z_3).$$

Take  $t^2_3$  to be the  $X$ -coordinate of a point in  $2E_3(\mathbb{Q})$ , with  $t_3 \neq x_3, y_3, z_3$ . For example (based on perusing tables of solutions),

$$t^2_3 = X(R_x + R_y) = \left( \frac{x_3y_0y_2 - x_0x_2y_3}{x^2_3 - y^2_3} \right)^2.$$

The condition  $t^2_3 - N_3 = t^2_0$  now takes the form

$$f_{31}f_{32}f_{33}f_{34} = 0, \quad (2.5)$$

where  $f_{3i}$  are irreducible polynomials in  $b, c, d, q, r, s$ , homogeneous of respective degrees 8, 8, 10, 10.

Similarly, on the curve

$$E_4 : Y^2 = X(X - N_4)(X - (N_4 - N_2)) = X(X - (x^2_4 - x^2_0))(X - (x^2_4 - x^2_2)),$$

with points

$$S_x(x^2_4, x_0x_2x_4), \quad S_y(y^2_4, y_0y_2y_4), \quad S_z(z^2_4, z_0z_2z_4),$$

take

$$t^2_4 = X(S_y - S_z) = \left( \frac{y_0y_2z_4 + y_4z_0z_2}{y^2_4 - z^2_4} \right)^2.$$

The condition  $t^2_4 - N_4 = t^2_0$  becomes

$$f_{41}f_{42}f_{43}f_{44} = 0, \quad (2.6)$$

where  $f_{4i}$  are irreducible polynomials in  $b, c, d, q, r, s$  of respective degrees 8, 8, 12, 12.

We can satisfy Eqs. (2.5), (2.6) by demanding  $f_{3i} = 0$ ,  $f_{4j} = 0$  for some  $1 \leq i, j \leq 4$ . By trial, it turns out that  $(i, j) = (4, 1)$  is a good choice. Consider the equations

$$f_{34} = 0, \quad f_{41} = 0,$$

namely

$$\begin{aligned} & b^2c^4q^3r - b^2c^3dq^3r - b^2c^2d^2q^3r + b^2cd^3q^3r - 2b^3c^3q^2r^2 - bc^5q^2r^2 + b^3c^2dq^2r^2 \\ & + 2bc^4dq^2r^2 + 2b^3cd^2q^2r^2 + bc^3d^2q^2r^2 - b^3d^3q^2r^2 - 2bc^2d^3q^2r^2 + b^4c^2qr^3 \\ & + 2b^2c^4qr^3 - 3b^2c^3dqr^3 - 3b^2c^2d^2qr^3 + 2b^2cd^3qr^3 + c^3d^3qr^3 - b^3c^3r^4 \end{aligned}$$

$$\begin{aligned}
& + b^3 c^2 d r^4 + b c^3 d^2 r^4 - b c^2 d^3 r^4 - b^2 c^4 q^3 s + b^2 c^3 d q^3 s + b^2 c^2 d^2 q^3 s - b^2 c d^3 q^3 s \\
& + 3 b^3 c^3 q^2 r s - 2 b^3 c^2 d q^2 r s - b c^4 d q^2 r s - b^3 c d^2 q^2 r s - 2 b c^3 d^2 q^2 r s + 3 b c^2 d^3 q^2 r s \\
& - 2 b^4 c^2 q r^2 s - b^2 c^4 q r^2 s - b^4 c d q r^2 s + 4 b^2 c^3 d q r^2 s + 4 b^2 c^2 d^2 q r^2 s - c^4 d^2 q r^2 s \\
& - b^2 c d^3 q r^2 s - 2 c^3 d^3 q r^2 s + b^3 c^3 r^3 s + b^3 c^2 d r^3 s - 2 b c^4 d r^3 s - 2 b^3 c d^2 r^3 s \\
& + b c^3 d^2 r^3 s + b c^2 d^3 r^3 s - 2 b^3 c^2 d q^2 s^2 + 2 b c^4 d q^2 s^2 + 2 b^3 c d^2 q^2 s^2 - 2 b c^3 d^2 q^2 s^2 \\
& + b^4 c^2 q r s^2 - b^2 c^4 q r s^2 + 2 b^4 c d q r s^2 - 2 b^2 c^3 d q r s^2 - 2 b^2 c^2 d^2 q r s^2 + 2 c^4 d^2 q r s^2 \\
& - b^2 c d^3 q r s^2 + c^3 d^3 q r s^2 - b^3 c^3 r^2 s^2 + b c^5 r^2 s^2 + b^3 d^3 r^2 s^2 - b c^2 d^3 r^2 s^2 \\
& - b^4 c d q s^3 + b^2 c^3 d q s^3 + b^2 c^2 d^2 q s^3 - c^4 d^2 q s^3 + b^3 c^2 d r s^3 - b c^4 d r s^3 \\
& - b^3 c d^2 r s^3 + b c^3 d^2 r s^3 = 0,
\end{aligned}$$

$$\begin{aligned}
& b^2 c^2 d q^2 r - b c^3 d q^2 r - b^2 d^3 q^2 r + b c d^3 q^2 r - 2 b^3 c d q r^2 + 2 b^2 c^2 d q r^2 + 2 b c d^3 q r^2 \\
& - 2 c^2 d^3 q r^2 + b^3 c d r^3 - b^2 c^2 d r^3 - b c d^3 r^3 + c^2 d^3 r^3 - b^2 c^3 q^2 s + b^2 c d^2 q^2 s \\
& + b c^2 d^2 q^2 s - b d^4 q^2 s + 2 b^3 c^2 q r s + 2 b^3 d^2 q r s - 4 b^2 c d^2 q r s - 4 b c^2 d^2 q r s \\
& + 2 c^3 d^2 q r s + 2 c d^4 q r s - 2 b^3 c^2 r^2 s + b^2 c^3 r^2 s - b^3 d^2 r^2 s + 2 b^2 c d^2 r^2 s \\
& + 2 b c^2 d^2 r^2 s - c^3 d^2 r^2 s + b d^4 r^2 s - 2 c d^4 r^2 s - 2 b^3 c d q s^2 + 2 b c^3 d q s^2 + 2 b^2 d^3 q s^2 \\
& - 2 c^2 d^3 q s^2 + 3 b^3 c d r s^2 - 2 b^2 c^2 d r s^2 - b c^3 d r s^2 - b^2 d^3 r s^2 - 2 b c d^3 r s^2 \\
& + 3 c^2 d^3 r s^2 - b^3 d^2 s^3 + b^2 c d^2 s^3 + b c^2 d^2 s^3 - c^3 d^2 s^3 = 0.
\end{aligned}$$

The second equation is quadratic in  $q$ , so it is necessary that the discriminant in  $q$  be square:

$$(b^2 - d^2)(dr - cs)(cr - ds)((bc + d^2)r - (b + c)ds)((b - c)dr - (bc - d^2)s) = u^2,$$

say. This represents the equation of an elliptic quartic  $C$  in  $r, s$  over  $\mathbb{Q}(b, c, d)$ , and a cubic model is

$$Y^2 = X(X + b^2(b^2 - d^2))(X - d^2(b^2 - d^2)),$$

with mappings

$$(X, Y) = \left( \frac{-b(b^2 - d^2)d((b - c)dr - (bc - d^2)s)}{(bc + d^2)r - (b + c)ds}, \frac{-b(b^4 - d^4)du}{((bc + d^2)r - (b + c)ds)^2} \right),$$

and inverse

$$\begin{aligned}
(r, s, u) = & (d(b(b^2 - d^2)(bc - d^2) + (b + c)X), b(b - c)(b - d)d^2(b + d) + (bc + d^2)X, \\
& -b(b^4 - d^4)(c^2 - d^2)^2 dY).
\end{aligned}$$

There is a point of infinite order  $Q(X, Y) = (b^2 d^2, b^3 d^3)$ . Now

$$Q + (-b^2(b^2 - d^2), 0) = \left( \frac{-d^4(b^2 - d^2)}{b^2}, \frac{-d^3(b^4 - d^4)}{b^3} \right),$$

and this latter point pulls back to the point on  $C$  with  $(r, s) = (d(b^3 - cd^2), (b^3c - d^4))$ . The equation  $f_{41} = 0$  now gives  $q = b(b - d)d(c + d)$ . Substitute

$$(q, r, s) = (b(b - d)d(c + d), d(b^3 - cd^2), b^3c - d^4) \quad (2.7)$$

into  $f_{34} = 0$ . Discarding trivial factors (factors that result in degenerate solutions to our problem), there results

$$\begin{aligned} & b^6 - b^5c - 2b^4c^2 - b^5d - 3b^4cd + b^3c^2d + b^2c^3d - b^4d^2 - b^3cd^2 + 3b^2c^2d^2 \\ & + bc^3d^2 + 2b^2cd^3 + bc^2d^3 - c^3d^3 = 0, \end{aligned}$$

and this latter equation defines a curve of genus 1. A cubic model is

$$E : Y^2 = X^3 + X^2 - 120X + 400,$$

with mappings

$$\begin{aligned} (X, Y) = & (4d(b^5 - 3b^4c + 4b^2c^3 - b^4d - b^3cd + 6b^2c^2d - b^3d^2 + b^2cd^2 + 4bc^2d^2 \\ & - c^3d^2 + 2bcd^3 - c^2d^3)/(c^2d^3(c + d)), \\ & - 4(8b^5c - 8b^4c^2 - 16b^3c^3 + 3b^5d - 11b^4cd - 26b^3c^2d + 4b^2c^3d - 3b^4d^2 \\ & - 17b^3cd^2 - 4b^2c^2d^2 + 12bc^3d^2 - 3b^3d^3 - 3b^2cd^3 + 18bc^2d^3 - 2c^3d^3 \\ & + 6bcd^4 + 2c^2d^4)/(c^2d^3(c + d))); \end{aligned}$$

and

$$\begin{aligned} b : c : d = & 2(80 - 40X + X^2)(40 - 10X + 2Y + XY) : \\ & (80 + 32X - 5X^2 - 12Y + XY)(40 - 10X + 2Y + XY) : \\ & 8(-4 + X)(5 + X)(80 - 40X + X^2). \end{aligned} \quad (2.8)$$

The torsion group on  $E$  is of order 2, generated by  $T_0 = (4, 0)$ , and the rational rank of  $E$  is one, with generator  $Q = (0, 20)$ . It is not difficult to verify now that we have a recipe for constructing infinitely many solutions to our original problem. The multiples of  $Q$  on  $E$  give rise via (2.8) to values of  $b, c, d$ , then via (2.7) to values of  $q, r, s$ , which in turn deliver  $(x_i, y_i, z_i, t_i)$ ,  $i = 0, \dots, 4$ . It is straightforward to check that points on  $E$  corresponding to degenerate solutions of the system (2.1) arise solely from  $x = -12, -5, 0, 4, 8, 20$ , corresponding to multiples of  $Q$  at most 3. Thus  $nQ$  for  $n \geq 4$  corresponds to a non-degenerate solution of our problem. We suppose now that  $mQ = (X_m, Y_m)$  and  $nQ = (X_n, Y_n)$ , where  $m > n \geq 4$ , return the same solution  $(x_0, y_0, z_0, t_0)$  (up to scaling). Equations (2.8), (2.7), and (2.2), (2.4), give  $(x_0, y_0, z_0, t_0)$  as functions of  $X_m, Y_m$  and of  $X_n, Y_n$ . It follows that, up to sign,  $x_0(X_m, Y_m) : y_0(X_m, Y_m)$  equals the ratio of two of  $x_0(X_n, Y_n)$ ,  $y_0(X_n, Y_n)$ ,  $z_0(X_n, Y_n)$ ,  $t_0(X_n, Y_n)$ . Suppose, for example,

$$x_0(X_m, Y_m) : y_0(X_m, Y_m) = x_0(X_n, Y_n) : y_0(X_n, Y_n). \quad (2.9)$$

Then

$$\begin{aligned}x_0(X_m, Y_m)y_0(X_n, Y_n) &= x_0(X_n, Y_n)y_0(X_m, Y_m), \\Y_m^2 &= X_m^3 + X_m^2 - 120X_m + 400, \\Y_n^2 &= X_n^3 + X_n^2 - 120X_n + 400.\end{aligned}\tag{2.10}$$

The system (2.10) defines a singular curve. Eliminating  $Y_m, Y_n$  results in an equation of the form

$$(X_m - X_n)g(X_m)g(X_n)h(X_m, X_n) = 0,$$

for polynomials  $g, h$ . By assumption,  $X_m \neq X_n$ . The rational roots of  $g(X) = 0$  occur at  $X = -12, -5, 0, 4, 8, 20$ , which are inadmissible values. Finally, the irreducible polynomial  $h$  defines a curve of genus 16 (as computed by Maple), so has only finitely many rational points.

For each choice of sign and for all choices of ratio on the right-hand side of (2.9), the genus of the corresponding irreducible curve, as computed by Maple, takes values only in the set  $\{16, 26, 36, 41, 56, 71\}$ , and is thus always greater than 1. Hence this irreducible curve has only finitely many rational points.

It follows that

$$x_0(X_n, Y_n) : y_0(X_n, Y_n) : z_0(X_n, Y_n) : t_0(X_n, Y_n)$$

as  $n \rightarrow \infty$  takes on infinitely many distinct values, as required.

### 3. Examples

Using the above construction, we find that the point  $3Q + T_0$  returns

$$\begin{aligned}(x_0, y_0, z_0, t_0) &= (41580, 225764, 455400, 693000), \\(x_1, y_1, z_1, t_1) &= (169323, 279125, 484077, 712173), \\(x_2, y_2, z_2, t_2) &= (309420, 380764, 549000, 757800), \\(x_3, y_3, z_3, t_3) &= (676500, 711964, 814440, 967560), \\(x_4, y_4, z_4, t_4) &= (854700, 883036, 967560, 1099560), \\(N_1, N_2, N_3, N_4) &= (26941381929, 94011840000, 455923353600, 728783193600);\end{aligned}$$

and the point  $4Q$  returns

$$\begin{aligned}(x_0, y_0, z_0, t_0) &= (720720, 7807800, 13993200, 24310440), \\(x_1, y_1, z_1, t_1) &= (2049905, 8040175, 14124175, 24386065),\end{aligned}$$



$$\begin{aligned}
 (x_2, y_2, z_2, t_2) &= (9724176, 12449976, 17024976, 26173224), \\
 (x_3, y_3, z_3, t_3) &= (17129970, 18811650, 22107150, 29730690), \\
 (x_4, y_4, z_4, t_4) &= (38591280, 39366600, 41043600, 45604440), \\
 (N_1, N_2, N_3, N_4) &= (3682673190625, 94040161560576, \\
 &\quad 292916434882500, 1488767454720000).
 \end{aligned}$$

#### 4. Concluding Remarks

Extension of the factor-difference problem to the case of five or more differences seems a very difficult problem. For the case of five differences, we must solve the system of 20 quadratic equations

$$(N_i =) x_i^2 - x_0^2 = y_i^2 - y_0^2 = z_i^2 - z_0^2 = t_i^2 - t_0^2 = w_i^2 - w_0^2, \quad i = 1, \dots, 5. \quad (4.1)$$

The values

$$\begin{aligned}
 (x_0, y_0, z_0, t_0, w_0) &= (400365, 1707240, 2583000, 4911144, 470278200/13), \\
 (x_1, y_1, z_1, t_1, w_1) &= (974925, 1924800, 2731680, 4990944, 470420160/13), \\
 (x_2, y_2, z_2, t_2, w_2) &= (1548365, 2269760, 2984800, 5133856, 470680000/13), \\
 (x_3, y_3, z_3, t_3, w_3) &= (3717675, 4071300, 4509180, 6146556, 2460\sqrt{36927467641}/13), \\
 (x_4, y_4, z_4, t_4, w_4) &= (4264533, 4576092, 4969692, 6491940, 3444\sqrt{18902740081}/13), \\
 (x_5, y_5, z_5, t_5, w_5) &= (15763475, 15850600, 15968680, 16505944, 512960840/13), \\
 (N_1, N_2, N_3, N_4, N_5) &= (790186622400, 2237142040000, 13660815272400, \\
 &\quad 18025949574864, 248326851942400)
 \end{aligned}$$

satisfy the system (4.1), but unfortunately  $w_3, w_4$  are not rational; thus only 18 of the 20 equations are satisfied.

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#### Appendix A

Table A.1 lists the first 30 solutions to the system (2.1) that we found by direct computation, ordered by  $N_1 + N_2 + N_3 + N_4$ . We list only  $(x_0, y_0, z_0, t_0)$  and  $(N_1, N_2, N_3, N_4)$ , with the  $x_i, y_i, z_i, t_i$  for  $i = 1, 2, 3, 4$  computable from (2.1).

Table A.1. Solutions to (2.1) ordered by  $N_1 + N_2 + N_3 + N_4$ .

| $(x_0, y_0, z_0, t_0)$        | $(N_1, N_2, N_3, N_4)$                           |
|-------------------------------|--------------------------------------------------|
| (216, 1116, 1656, 4724)       | (274833, 1225728, 3368448, 30159360)             |
| (872, 4328, 5003, 11672)      | (2062016, 14170112, 24040016, 78094016)          |
| (1251, 4149, 8501, 26595)     | (5595975, 8643024, 14595399, 200643399)          |
| (504, 2556, 3144, 10116)      | (5348673, 23016960, 64903168, 207884313)         |
| (420, 2595, 3684, 4380)       | (7441200, 20254000, 53406000, 262652544)         |
| (1512, 2580, 9720, 13380)     | (30067200, 64054881, 146236825, 269568000)       |
| (1024, 6865, 11360, 21440)    | (12084800, 25206800, 322088000, 897512000)       |
| (63, 8055, 16695, 37695)      | (26128575, 291722431, 561117375, 713526975)      |
| (1188, 6588, 11188, 38412)    | (22208256, 53719281, 456120756, 1890838656)      |
| (1320, 3828, 8520, 18920)     | (6690816, 58397625, 182131200, 2383603200)       |
| (4840, 12440, 38765, 63592)   | (14766800, 660920000, 1081472000, 1242510800)    |
| (525, 9485, 10515, 86765)     | (6074775, 56154519, 562837275, 2551994775)       |
| (312, 3237, 17112, 78312)     | (32621056, 73519056, 703213056, 2438287056)      |
| (4760, 13860, 22792, 118440)  | (56908800, 481614336, 1089898425, 2060352000)    |
| (4472, 6292, 9136, 29744)     | (21063680, 73258865, 128061440, 3912796160)      |
| (2955, 22920, 35160, 92616)   | (49714000, 280098000, 707526144, 3658176000)     |
| (2520, 9765, 10680, 39240)    | (71442000, 357696000, 835810000, 4295960064)     |
| (630, 10890, 32970, 256410)   | (64631196, 1135630125, 1625305500, 5397737884)   |
| (6416, 23184, 70056, 121464)  | (112446180, 389521953, 2462035968, 5727541248)   |
| (5145, 17241, 44345, 154455)  | (97405875, 969751944, 1548825075, 8252709075)    |
| (2688, 19712, 29337, 68838)   | (151534656, 522510912, 3738214656, 7048774656)   |
| (6405, 21987, 44163, 56763)   | (56669431, 589789431, 2630211831, 9997212456)    |
| (9636, 13233, 18788, 64284)   | (63497520, 258485040, 1292773680, 12704186880)   |
| (7880, 21560, 37345, 43960)   | (132675536, 1074944000, 4126584000, 10737272000) |
| (13860, 28105, 34020, 58212)  | (616734000, 794865456, 4494672000, 11811294000)  |
| (3465, 8316, 33660, 47740)    | (272354544, 671554800, 2553922800, 15261246000)  |
| (10592, 15533, 18208, 54208)  | (55771136, 504835136, 1615043136, 16824229136)   |
| (6720, 37968, 64680, 139920)  | (206222625, 677376000, 2019635200, 16670745700)  |
| (4700, 11925, 36900, 91620)   | (624696624, 1645815600, 4629150000, 15904350000) |
| (15120, 56700, 91448, 134120) | (220359321, 671385600, 7094466225, 15647385600)  |

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